

Biebele Abel, M.S.

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EDUCATION

- 2015 Ph.D., Bioenvironmental Science, Morgan State University, Baltimore, MD
 Dissertation: Enzymatic Signal Amplification Using Plasmonic Nanostructures for Applications in Bioenvironmental Science and Engineering.
 Mentor: Dr. Kadir Aslan
- 2011 M.S., Science (Analytical Chemistry), Morgan State University, Baltimore, MD
- 2006 B.Sc., Chemistry, University of Maiduguri, Borno State, Nigeria.

RESEARCH INTERESTS

Nanotechnology: Plasmonics, Metal-Enhanced Fluorescence, Metallic Nanoparticles, Ultra-Fast Nanoparticles Based Assay.

Biotechnology: Biosensor, Plasmon-Enhanced Enzymatic Reaction.

Dissertation Research (2011-Present): *Enzymatic Signal Amplification Using Plasmonic Nanostructures for Applications in Bioenvironmental Science and Engineering.*

We investigated the fundamental study of the interactions of plasmonic nanoparticles with enzymes (i.e., hybrid systems) that led to the improvement of the stability and increased efficiency of enzymes used in bioenvironmental science and engineering. The hybrid system, which is comprised of a solid platform (glass, electrodes, and high-throughput screening microplates), plasmonic nanoparticles (silver, gold, copper, and nickel), immobilizers or linkers (biotinylated proteins and functional linkers) and enzymes (horse radish peroxidase, alkaline phosphatase, and β -galactosidase), was prepared in a facile and inexpensive manner that employed commercially available enzymes.

RESEARCH EXPERIENCE

2011 – Present: *Graduate Research Assistant*

Designed and conducted research on the enhancement of enzymatic colorimetric response using nanostructured materials. Mentor- Dr. Kadir Aslan

Conducted research on understanding the requirements for the use of plasmonic nanoparticles in combination with enzymes that can be readily incorporated into biosensing and other applications.

2009 – 2010: *Research Assistant*

Designed and conducted research on the characterization of chlorophenols using fluorescence and electrochemical methodology. Mentor- Dr. Maurice Iwunze

Conducted research on the characterization of chlorophenols (i.e., 4-chlorophenol, 2,5-dichlorophenol, 2,4,5-trichlorophenol and Pentachlorophenol), which was achieved by correlating the half-wave potential, $E_{1/2}$, obtained using Cyclic Voltammetry (CV) with the fluorescence emission energy, E_f , obtained by steady-state fluorescence.

2010: *Summer Research internship*

NASA Goddard Space Flight Center, Greenbelt, MD.

Analyzed data obtained from satellite (and specifically Moderate Resolution Imaging Spectroradiometer (MODIS) fire radiative energy measurements for use in emissions estimation. Mentor-Dr. Charles Ichoku

TEACHING

Morgan State University

Course Instructor

Principles of General Chemistry I- Laboratory (CHEM 105L), Spring 2011- Fall 2013

Prepared and supervised laboratory techniques; taught about 20-30 students; set up exam questions, and maintained a database of student grades on Blackboard.

Teaching Assistant

Principles of General Chemistry II-Laboratory (CHEM 106L), Spring 2010

PREVIOUS EXPERIENCE

Tutor

Leigh Walk Elementary School, Baltimore MD. Community Outreach, Morgan State University Chemistry Club (Spring 2009). Tutored 3rd and 4th grade students; engaged students in fun science activities and laboratory experiments.

PUBLICATIONS

Abel, B., Coskun, S., Mohammed, M., Williams, R., Unalan*, H.U., Aslan*, K. "Metal-Enhanced Fluorescence from Silver Nanowires with High Aspect Ratio", *The Journal of Physical Chemistry C*, (2015), 119(1), 675-684.

Abel, B., Kabir, T., Odukoya, B., Mohammed, M., Aslan*, K. "Enhancement of Colorimetric Response of Enzymatic Reactions by Thermally Evaporated Plasmonic Thin Films: Application to Glial Fibrillary Acidic Protein", *Analytical Methods*, (2015), 7, 1175-1185.

Kioko, B., Ogundolie, T., Adebisi, M., Ettinoffe, Y., Rhodes, C., Gordon, B., Thompson, N., Mohammed, M., **Abel, B.**, Aslan*, K. "De-crystallization of uric acid crystals in synovial fluid using gold colloids and microwave heating", *Nano Biomedicine and Engineering*, (2014), 6(4), 104-110.

Abel, B. and Aslan, K. "Immobilization of enzymes to silver island films for enhanced enzymatic activity", *Journal of colloid and interface science* (2014), 415, 133-142.

Abel, B., Clement, T. C., Aslan, K*. "Enhancement of Enzymatic Colorimetric Response by Silver Island Films on High Throughput Screening Microplates", *Journal of Immunological Methods* (2014), 411, 43-49.

Abel, B. and Aslan, K*. "Plasmon-Enhanced Enzymatic Reactions 2: "Optimization of Enzyme Activity by Surface Modification of Silver Island Films with Biotin-Poly(Ethylene-glycol)-Amine", *Nano Biomedicine and Engineering* (2012), 4(1), 23-28.

Abel, B. and Aslan, K*. "Surface modification of plasmonic nanostructured materials with thiolated oligonucleotides in 10 seconds using selective microwave heating", *Annalen der Physik* (2012), 542(11), 741-750.

Iwunze, M. O. and **Abel, B.** "Determination of Half-wave Potentials of Selected Chlorophenols", *Portugaliae Electrochimica Acta* (2012), 30(5), 329-334.

Abel, B., Akinsule, A., Andrews, C., Aslan, K*. "Plasmon-Enhanced Enzymatic Reactions Study of Nanoparticle-Enzyme Distance- and Nanoparticle Loading-Dependent Enzymatic Activity", *Nano Biomedicine and Engineering* (2011), 3(3), 184-191.

CONFERENCE PRESENTATIONS

"Plasmon-Enhanced Enzymatic Reaction: Application to β -Galactosidase and Alkaline Phosphatase." Paper presentation at the American Chemical Society (ACS) 245th National Meeting and Exposition in New Orleans, LA, April 2013.

"Surface Modification of Plasmonic Nanostructured Materials with Thiolated Oligonucleotides in 10 seconds using Selective Microwave Heating." Paper presentation at the National Organization for the Professional Advancement of Black Chemist and Chemical Engineers (NOBCCChE) 39th annual meeting in Washington, DC, April 2012.

"Plasmon-Enhanced Enzymatic Reaction: Optimization of Enzyme Activity by Surface Modification of SIFs with biotinylated polyethylene glycol diamine." Paper presentation at the Minority Serving Institutions Research Partnership Consortium in Baltimore, MD, March 2012.

"The Correlation of Voltammetric Half-Wave Potential and the Fluorescence of Phenol and Selected Chlorophenols in CTAB Surfactant Solution." Paper presented at the Minority Serving Institutions Research Partnership Consortium in Baltimore, MD, Sept 2010.

"Determination of Chlorophenols in Solution, Using Copper (II) Complex as a Chemical Sensor." Poster presentation at the annual conference of the National Organization for the Professional Advancement of Black Chemist and Chemical Engineers (NOBCCChE) in Baltimore, MD, Sept 2009.

"Satellite Observation of Biomass Burning Emission." Summer interns poster presentation at NASA Goddard Space Flight Center in Greenbelt, MD, August 2010.

PROFESSIONAL MEMBERSHIPS

2008- Present: Member, American Chemical Society (ACS)

2008- Present: Member, National Organization for Advancement of Black Chemist and Chemical Engineers (NOBCCChE)

SOFTWARE PACKAGES: Sigma Plot®, PSI Plot®, EndNote®, Chemical Software (Chem draw).

INSTRUMENTAL SKILLS: Ultra Visible spectroscopy, Infra-Red spectroscopy, Atomic Absorption spectroscopy, Gas Chromatography, Mass Spectroscopy, Fluorescence Spectroscopy, Electrochemical Instrument (Potentiostat).

LANGUAGES SPOKEN: Fluent in English and Nigerian Pidgin English.